

The trace-arithmetic distance is a metric on the full rank-one positive cone

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Status and source question

Status: candidate partial result, likely valid.

For $p > 0$ and positive semidefinite matrices A, B , Zhang [1] uses the operations

$$A \kappa_p B = \left(A^{p/4} B^{p/2} A^{p/4} \right)^{1/p}, \quad d_p(A, B)^2 = \text{Tr} \left(\frac{A + B}{2} - A \kappa_p B \right).$$

Problem 1.2 on page 2 of the source asks whether d_p is a metric on $\mathbb{M}_n(\mathbb{C})_+$ for every $1 < p < 2$, and asks the corresponding question for a general C^* -algebra with a faithful trace. Here is the source statement.

Problem 1.2 ([6] Problem 2). *Let $n \geq 2$. Is d_p^{Tr} on $\mathbb{M}_n(\mathbb{C})_+$ a metric for each $1 < p < 2$, and is it not a metric for each $0 < p < 1$? More generally, if $\mathcal{A}$ is a C^* -algebra with a faithful trace τ , is d_p^τ a metric for $1 < p < 2$?*

The source gives a counterexample at $p = 1/2$ but states that the full-cone case $1 < p < 2$ remains open [1, Remark 4.3]. The result below proves the metric property, throughout the entire requested parameter interval, on the cone consisting of all positive multiples of rank-one projections, with the zero matrix included. In fact the proof works for every $0 < p \leq 2$.

Proof intuition

Write a rank-one positive matrix as aP , where $a \geq 0$ and P is a rank-one projection. If $P = uu^*$ and $Q = vv^*$, the trace of the noncommutative mean collapses to

$$\text{Tr}(aP \kappa_p bQ) = \sqrt{ab} |\langle u, v \rangle|^{2/p}.$$

Thus the only directional datum is the Fubini–Study angle $\theta(P, Q) = \arccos |\langle u, v \rangle|$. For $\alpha = 2/p \geq 1$, deform this angle by

$$f_\alpha(t) = \arccos((\cos t)^\alpha).$$

The key one-variable fact is that f_α is increasing and concave on $[0, \pi/2]$, hence it turns the projective angle into another metric. After putting $r = \sqrt{a}$ and $s = \sqrt{b}$, the distance becomes

$$\sqrt{2} d_p(aP, bQ) = \sqrt{r^2 + s^2 - 2rs \cos(f_\alpha(\theta(P, Q)))}.$$

This is exactly the Euclidean cone metric over the deformed projective metric. The triangle inequality then follows by realizing any three base distances as the angular distances of three unit

vectors and applying the ordinary Euclidean triangle inequality. This argument controls the radial variables a, b that are absent from the previously known fixed-trace projector result.

The same radial decomposition also gives an exact reformulation of the full problem. On density matrices put $F_p(\rho, \sigma) = \text{Tr}(\rho \kappa_p \sigma)$ and $\delta_p = \arccos F_p$. Then d_p is a metric on the full positive cone if and only if δ_p is a metric on density matrices. Equivalently, every three-state matrix $[F_p(\rho_i, \rho_j)]_{i,j=1}^3$ must be positive semidefinite. This isolates the unresolved part as one explicit trace inequality; the packet does not prove that inequality, so the full cone remains open.

Formal proof

Let

$$\mathcal{R}_n^+ := \{0\} \cup \{aP : a > 0, P \in \mathbb{M}_n(\mathbb{C})_+, P^2 = P, \text{rank } P = 1\}.$$

Lemma 1 (Projective-angle deformation). *Let $\theta(P, Q) = \arccos |\langle u, v \rangle|$ for $P = uu^*$ and $Q = vv^*$, where u, v are unit vectors. If $\alpha \geq 1$ and*

$$f_\alpha(t) = \arccos((\cos t)^\alpha), \quad 0 \leq t \leq \frac{\pi}{2},$$

then

$$\delta_\alpha(P, Q) := f_\alpha(\theta(P, Q))$$

is a metric on the rank-one projective space and takes values in $[0, \pi/2]$.

Proof. The Fubini–Study angle θ is a metric. For completeness, this follows from the spherical triangle inequality: choose phases of successive unit representatives so that the spherical angles of the first and second, and of the second and third, are respectively $\theta(P, Q)$ and $\theta(Q, R)$. The spherical angle between the first and third representatives is at least the minimum over their relative phase, which is $\theta(P, R)$, and is at most the sum of the other two angles.

It remains to record the properties of f_α . It is increasing, $f_\alpha(0) = 0$, and $f_\alpha(t) \geq t$, because $(\cos t)^\alpha \leq \cos t$. We now prove concavity. On $(0, \pi/2)$,

$$f'_\alpha(t) = \frac{\alpha(\cos t)^{\alpha-1} \sin t}{\sqrt{1 - (\cos t)^{2\alpha}}}.$$

With $y = \cos^2 t$, the square of $f'_\alpha(t)/\alpha$ is

$$G(y) = \frac{y^{\alpha-1}(1-y)}{1-y^\alpha}.$$

After multiplication by the positive denominator, the sign of $G'(y)$ is the sign of

$$h(y) = \alpha - 1 - \alpha y + y^\alpha.$$

Since $h(1) = 0$ and $h'(y) = \alpha(y^{\alpha-1} - 1) \leq 0$ on $[0, 1]$, one has $h(y) \geq 0$. Hence G is increasing in y . Since $y = \cos^2 t$ decreases with t , f'_α decreases, so f_α is concave on $[0, \pi/2]$.

Concavity and $f_\alpha(0) = 0$ imply subadditivity whenever $x, y, x+y \in [0, \pi/2]$: indeed,

$$f_\alpha(x) \geq \frac{x}{x+y} f_\alpha(x+y), \quad f_\alpha(y) \geq \frac{y}{x+y} f_\alpha(x+y).$$

For three projections put $x = \theta(P, Q)$ and $y = \theta(Q, R)$. If $x + y \leq \pi/2$, monotonicity, the triangle inequality for θ , and subadditivity give

$$\delta_\alpha(P, R) \leq f_\alpha(x + y) \leq f_\alpha(x) + f_\alpha(y).$$

If $x + y > \pi/2$, then

$$\delta_\alpha(P, R) \leq \frac{\pi}{2} < x + y \leq f_\alpha(x) + f_\alpha(y).$$

Definiteness and symmetry are immediate. Thus δ_α is a metric. $\square$

Lemma 2 (Elementary metric-cone lemma). *Let (X, δ) be a metric space with $0 \leq \delta \leq \pi/2$. Identify all points $(0, x)$ in $[0, \infty) \times X$, and define*

$$D((r, x), (s, y))^2 = r^2 + s^2 - 2rs \cos \delta(x, y).$$

Then D is a metric on the resulting cone.

Proof. Only the triangle inequality needs proof. Given $x, y, z \in X$, put

$$a = \delta(x, y), \quad b = \delta(y, z), \quad c = \delta(x, z).$$

The metric inequalities give $|a - b| \leq c \leq a + b$, and $a + b \leq \pi$. Consequently there are unit vectors $U, V, W \in \mathbb{R}^3$ whose pairwise angles are a, b, c . Explicitly, when $\sin a \sin b \neq 0$, take

$$V = (1, 0, 0), \quad U = (\cos a, \sin a, 0),$$

$$W = (\cos b, \sin b \cos \gamma, \sin b \sin \gamma), \quad \cos \gamma = \frac{\cos c - \cos a \cos b}{\sin a \sin b}.$$

The inequalities $|a - b| \leq c \leq a + b$ say exactly that $|\cos \gamma| \leq 1$; the cases with a zero sine follow by a limit or directly. For nonnegative r, s, t , the Euclidean triangle inequality now gives

$$\|rU - tW\|_2 \leq \|rU - sV\|_2 + \|sV - tW\|_2,$$

which is precisely

$$D((r, x), (t, z)) \leq D((r, x), (s, y)) + D((s, y), (t, z)).$$

Symmetry and nonnegativity are clear. Equality $D = 0$ identifies exactly the same positive-radius base point, together with the already identified zero-radius points. $\square$

Theorem 1 (Exact reduction of the full-cone problem). *Fix $n \geq 2$ and a value of p for which d_p is a definite nonnegative function. Let*

$$\mathcal{D}_n = \{\rho \in \mathbb{M}_n(\mathbb{C})_+ : \text{Tr } \rho = 1\}, \quad F_p(\rho, \sigma) = \text{Tr}(\rho \kappa_p \sigma),$$

and set $\delta_p(\rho, \sigma) = \arccos F_p(\rho, \sigma)$. The following are equivalent:

1. d_p is a metric on $\mathbb{M}_n(\mathbb{C})_+$;
2. δ_p is a metric on $\mathcal{D}_n$;
3. for all $\rho, \sigma, \omega \in \mathcal{D}_n$,

$$1 + 2F_p(\rho, \sigma)F_p(\sigma, \omega)F_p(\omega, \rho) \geq F_p(\rho, \sigma)^2 + F_p(\sigma, \omega)^2 + F_p(\omega, \rho)^2. \quad (1)$$

For $1 < p < 2$, therefore, the matrix part of Problem 1.2 is exactly the three-state trace inequality (1).

Proof. The scalar overlap is symmetric because

$$F_p(\rho, \sigma) = \text{Tr} |\sigma^{p/4} \rho^{p/4}|^{2/p},$$

and an operator and its adjoint have the same singular values. The assumed nonnegativity and definiteness of d_p give $0 \leq F_p(\rho, \sigma) \leq 1$, with equality at the upper endpoint exactly when $\rho = \sigma$. Thus δ_p is a well-defined symmetric definite function with values in $[0, \pi/2]$.

For nonzero A, B , write

$$A = a\rho, \quad B = b\sigma, \quad a = \text{Tr } A, \quad b = \text{Tr } B, \quad \rho, \sigma \in \mathcal{D}_n.$$

Separate homogeneity of the mean gives

$$\text{Tr}(A \kappa_p B) = \sqrt{ab} F_p(\rho, \sigma),$$

and hence

$$2d_p(A, B)^2 = a + b - 2\sqrt{ab} \cos \delta_p(\rho, \sigma). \quad (2)$$

Thus $\sqrt{2}d_p$ is precisely the cosine-cone distance with radial coordinate $\sqrt{\text{Tr } A}$. Lemma 2 proves that (2) implies (1).

Conversely, suppose that the triangle inequality for δ_p fails. After relabeling, there are density matrices ρ, σ, ω such that

$$\gamma := \delta_p(\rho, \omega) > \alpha + \beta, \quad \alpha := \delta_p(\rho, \sigma), \quad \beta := \delta_p(\sigma, \omega).$$

All these angles lie in $[0, \pi/2]$, and strict failure forces $\alpha, \beta > 0$. In the Euclidean plane let X, Y, Z be unit vectors on rays of angles $0, \alpha, \alpha + \beta$, respectively. The line segment from X to Z meets the intermediate ray at sY for some $s > 0$. Apply (2) to

$$A = \rho, \quad B = s^2\sigma, \quad C = \omega.$$

Because sY lies on the segment $[X, Z]$,

$$\begin{aligned} \sqrt{2}(d_p(A, B) + d_p(B, C)) &= \|X - sY\|_2 + \|sY - Z\|_2 \\ &= \|X - Z\|_2 = \sqrt{2 - 2\cos(\alpha + \beta)} \\ &< \sqrt{2 - 2\cos \gamma} = \sqrt{2}d_p(A, C). \end{aligned}$$

Thus failure of the angular triangle inequality forces failure on the positive cone, proving (1) implies (2).

It remains to identify (2) with (3). Put

$$x = F_p(\rho, \sigma) = \cos \alpha, \quad y = F_p(\sigma, \omega) = \cos \beta, \quad z = F_p(\omega, \rho) = \cos \gamma.$$

The left side minus the right side of (1) is the determinant of the symmetric matrix with diagonal entries 1 and off-diagonal entries x, y, z . Moreover,

$$1 + 2xyz - x^2 - y^2 - z^2 = \sin^2 \alpha \sin^2 \beta - (\cos \gamma - \cos \alpha \cos \beta)^2.$$

Since $\alpha, \beta, \gamma \in [0, \pi/2]$, nonnegativity is equivalent to $|\alpha - \beta| \leq \gamma \leq \alpha + \beta$, exactly the three triangle inequalities. This proves the remaining equivalence. $\square$

Theorem 2 (Metricity on the full rank-one positive cone). *For every $n \geq 2$ and every $0 < p \leq 2$, the restriction of d_p to $\mathcal{R}_n^+$ is a metric. In particular, this holds for every $1 < p < 2$, the parameter range left open on the full cone in Problem 1.2 of [1].*

Proof. Let $A = aP$ and $B = bQ$, with $a, b > 0$, $P = uu^*$, $Q = vv^*$, and u, v unit vectors. Functional calculus and $PQP = |\langle u, v \rangle|^2 P$ yield

$$A^{p/4} B^{p/2} A^{p/4} = (ab)^{p/2} |\langle u, v \rangle|^2 P.$$

Therefore

$$\text{Tr}(A \kappa_p B) = \sqrt{ab} |\langle u, v \rangle|^{2/p}.$$

The same formula, by continuity, also covers either $a = 0$ or $b = 0$. Put

$$r = \sqrt{a}, \quad s = \sqrt{b}, \quad \alpha = \frac{2}{p} \geq 1,$$

and let $\theta(P, Q) = \arccos |\langle u, v \rangle|$. Then

$$\begin{aligned} 2d_p(aP, bQ)^2 &= r^2 + s^2 - 2rs(\cos \theta(P, Q))^\alpha \\ &= r^2 + s^2 - 2rs \cos \delta_\alpha(P, Q). \end{aligned}$$

By Lemma 1, δ_α is a metric with diameter at most $\pi/2$. Lemma 2 therefore says that the square root of the last expression is a metric on the cone. Since this square root equals $\sqrt{2} d_p$, the same is true of d_p . The cone point at radius zero is exactly the zero matrix, so definiteness has the intended matrix meaning. $\square$

Verification, scope, and novelty check

The proofs are analytic; the script `code/check_rank_one_metric.py` only checks the trace formula and random triangle instances and is not part of the proof. The crucial reviewer checks are the sign calculation proving concavity of f_α , the use of the elementary metric-cone lemma, and both directions of Theorem 1.

This result does *not* settle Problem 1.2 on the full matrix cone or for a general C^* -algebra. The noncommutative difficulty begins once matrices have rank at least two. Theorem 1 isolates that difficulty as the three-state Gram determinant inequality (1), which is not proved here.

A bounded novelty search was performed before promotion:

- the local run indexes and parsed source collection were searched by arXiv id; related searches used rank-one matrices, projectors, trace arithmetic, and the symmetric Bures–Wasserstein divergence;
- Komálovics–Molnár [2] prove the threshold only on the normalized rank-one projector set;
- a targeted web/full-text check of Vuong’s April 2026 follow-up [3] found the commuting cone and the normalized rank-one projector set, and explicitly says that full-cone metricity for the corresponding parameters remains open;
- no explicit theorem for arbitrary scalar multiples aP was found.

The extension may be regarded as a natural metric-cone completion of the known projector theorem, so novelty confidence is moderate rather than high.

Human-review recommendation. Promote after checking Lemma 1, especially the monotonicity direction in the $G(y)$ calculation, and confirming that the searched literature does not use “rank-one cone” to include arbitrary non-normalized scalar multiples.

References

- [1] T. Zhang, *Trace arithmetic- κ_p inequality*, arXiv:2602.11922, 2026.
- [2] Á. Komálovics and L. Molnár, *On a parametric family of distance measures that includes the Hellinger and the Bures distances*, J. Math. Anal. Appl. 529 (2024), no. 2, article 127226.
- [3] T.-D. Vuong, *Quantitative geometry of the symmetric α -z Bures–Wasserstein divergence*, Filomat, prepublication manuscript, April 2026.